NPLify · P0 Technical Baseline

Test Strategy & Coverage

New XP Technologies Limited
21 October 2026 · Confidential · new document, added for the design review (Pete, points 6a to 6d; Andrew, items 2.5 and 9.4)

1 · What this document is for

The pack already contains the yardsticks: the acceptance vectors of the Calculation Specification, the 61 worked transactions of the Ledger Posting Design, the state machines of the Lifecycle & Approvals document, the field groups of the Roles & Visibility Matrix, the constraints of the ERD's Appendix C and the exit criteria of the Cutover Approach. It did not say in one place which tests exist, what each proves, where it runs and what passing means. This document does. It is the testing framework Pete asked for: full scope, the purpose and expected result of every test, regression so the system stays robust as the project evolves, and a map from the Workflow Tree to the tests, so that well-tested and under-tested areas are visible.

2 · Test levels and where each runs

Level	What it proves	Where it runs	Who writes it	When
unit	One function gives the expected figure: pricing, rounding, breach check, tier split	in the build, on every change	Engineering, from the vectors	from M0
fixture	A posted event produces exactly the lines of a worked transaction, and the identities hold afterwards	in the build, against a fresh database	Engineering, from T1 to T61	from M2
database	A write that breaks a constraint is refused by the database itself, not only by the application	in the build, against the migrated schema	Engineering, from Appendix C	from M2
integration	A sequence of actions through the API follows the state machine, the approvals and the visibility rules	in the build and on staging	Engineering, from the Lifecycle and Roles documents	M3 onward
release	A release can be applied and rolled back; migrations run against last week's snapshot with the ledger checks green	staging, before every release	Engineering	every release
UAT	NPL staff run real procedures with real evidence and get the result the SOP says	staging	NPL Operations and Finance, with New XP	M10
parallel run	The app reproduces the sheets, day after day, to the cent	staging	NPL Finance	M10, three to four weeks
acceptance	The go-live criteria 1 to 13 of the Cutover Approach	staging, then production on the switch day	Engineering with NPL sign-off	M10, M11

3 · The acceptance gate per phase, and the regression rule

Every feature moves through the proposal's pipeline (mockup, build, QA, staging, UAT, production). Each phase closes with a demonstration against its gate; the gate is the set of tests below, all green. The regression rule: every release, in every phase and under the support bundle after go-live, runs the full vector, fixture, state

and visibility suites, and a failing exact test blocks the release. A test is never removed because it fails; it is fixed or its expected result is changed with a recorded reason and NPL's agreement where the figure is NPL's.

Phase gate	Must be green
Phase 1 (M0 to M5): the calculating core	V-L1…L13, V-B1…B23, V-RB1…3, V-ED-A…D, V-TIER1…2, F-T1…T61 as far as the patterns are built, F-ID1…12, F-REPLAY, F-REV, D-CONS1…28, D-POLY1…7, S-DEAL, S-SETT, S-RATE, S-RR, C-APPR, C-JOBS; the end-to-end demonstration reproduces the reference workbook figures
Phase 2 (M6, M7): hardening	everything above plus R-VIS1…n, R-ACCESS, C-LOCK, S-EMERG, C-INT1…3 on staging; the role-gated, reconciled platform with side-by-side comparison reports
Phase 3 (M8 to M11): completion and go-live	everything above plus O-EVID1…3, O-C6A, O-SLA, G-SEC, G-CONC, G-OUT, G-DUP, G-REST, G-OPEN, G-FILTER, G-RECON, G-VULN, D-MIG; the go/no-go on Finance's discrepancy log
After go-live (support bundle)	the full suite on every release; G-REST rehearsed once a year; the coverage sheet reviewed at every quarterly access review

4 · Identifiers and result classes

Every test has an identifier that never changes and is never reused (the same rule as the T, L, B, W, F and C identifiers of the pack; the Reading Companion's register lists them). The first letter says the family: V vectors and calculation, F fixtures and identities, S state machines, C concurrency and controls, D database, R roles and visibility, O operating procedures and UAT, G go-live acceptance. A purpose line says what the test proves. A result class says what passing means, so that a reviewer can tell a test with one right answer from a test whose result must only be reasonable:

Result class	Passing means	Examples
exact	The result equals the expected value to the cent (eight decimals for BTC amounts, the unit where NPL's sheet rounds), or the expected state or refusal occurs	V-L1, F-T2, S-DEAL, D-CONS
tolerance	The result is within the rounding of the pair (a dropped fraction or a round-up with its owner)	cash rounding cases; margin-of-the-day reconciliation
range	The result falls inside a configured band, because the right answer depends on data or judgement	G-RECON checks with a tolerance; G-VULN severity; aging thresholds
invariant	A property that must always be true holds after every step and every fixture	F-ID1…12, F-REPLAY, C-APPR, R-VIS, G-DUP

5 · Test inventory

One row per test or test family; the family's members are numbered in the test suite itself (vectors-v5.9.csv for V, the fixtures for F). The source column names where the expected result is written down in the pack; the phase column when the test first has to be green.

Id	Purpose	Result class	Level	Source in the pack	Phase
V-L1 … V-L13	The pricing function reproduces each acceptance vector A: sender rate, parts, client net, delivered, earnings, payable, residual	exact (to the cent; BTC to 8 decimals)	unit	Calculation Specification §18, 18a; vectors-v5.9.csv	M1, M4
V-B1 … V-B23	The pricing function reproduces NPL's own figures: the sheet lines B1 to B18 (B2 and B18 against the corrected sheet of 21 October) and the five figures B19 to B23 NPL supplied on 21 October for the 18a cases and the same-currency deal	exact (to the unit where the sheet rounds)	unit	Calculation Specification §19, 19a, 19b	M1, M4

V-RB1 … V-RB3	Breach rules 2a: thin history → fallback threshold; zero or negative baseline → no percentage formula; absolute tolerance either way → flagged, not frozen	exact (breach / flag / neither)	unit	Calculation Specification 2a; THRESHOLD kinds	M1
V-ED-A … V-ED-D	Effective-date scenarios A to D: a fee change on 1 November applies only to deals priced from then; fees, tiers, rounding and partner settings	exact	integration	Configuration Schema 1a; Lifecycle 2a	M4
V-TIER1, V-TIER2	Tier exactly reached and tier crossed on GDC (L12, L13); a void lowers the counter; two simultaneous quotes read the counter in order	exact	integration	Calculation Specification 5a; ERD Appendix D	M0, M3
F-T1 … F-T61	Each worked transaction posts exactly the lines, currencies, ownership tags and components of the Ledger Posting Design, and the balances after it	exact	fixture	Ledger Posting Design T1 to T61; Reading Companion register	M2
F-ID1 … F-ID12	The identities of Section 20 hold after every fixture: client assets = liabilities per currency; PAYABLE = Σ GROUP_ENTITLEMENT; partner balance = client + pool; rebates and shares agree to accruals; losses close; own wallets hold client money in four cases only; every released line is accounted for	invariant	fixture	Ledger Posting Design §20	M2
F-REPLAY	A retried event with the same idempotency key posts nothing (rule 1a), for every source type of §22	invariant	integration	Ledger Posting Design 1a; Architecture §5	M2
F-REV	A correction is a reversal naming what it reverses plus a re-posting, dated today with corrects_period (rule 11a); a reported month never changes	invariant	integration	Ledger Posting Design 11a, Pattern E	M2
S-DEAL	Every transition of Figure 1 with its guard and its approver; every refused transition (wrong state, wrong role, same person) refused	exact	integration	Lifecycle §2, 2a, 2b, 7, 7a	M3 to M5
S-SETT	Settlement run and lines: approve, release, confirm full and short, return; "sent, not yet confirmed" shown and aged; after release nothing is cancelled	exact	integration	Lifecycle §3, Figure 2; Ledger rule 13, 13a	M5
S-RATE	Rate version states: breach freezes the pair; Finance approval with a reason; expiry	exact	integration	Lifecycle Figure 3	M1
S-RR	Reroute before and after conversion: custody bounded, forward, dues recovered, Management ratification for post_conversion	exact	integration	Lifecycle Figure 4, 2b; Pattern O	M5
S-EMERG	Emergency suspend / freeze / cutoff override takes effect at once, blocks what it must, is lifted if not ratified within a business day	exact	integration	Lifecycle §5 addition; ERD EMERGENCY_ACTION	M6
C-APPR	Approval valid only for the record version seen; two approvers at once: one wins; approver's person_ref never the initiator's	invariant	integration	Lifecycle 6a; Roles 5a	M3 to M6
C-JOBS	A job started twice runs once; catch-up after a restart; bounded retries; the job-health line	invariant	integration	Architecture §4 addition	M1 onward
C-LOCK	A record lock is released on save or cancel, expires after lock_idle_minutes, can be cleared by Management, and every step is in AUDIT_LOG	exact	integration	Lifecycle 6a; ERD RECORD_LOCK	M6

C-INT1 … C-INT3	Interruption cases of Pattern P: partner executed and the application failed; confirmation never arrives; explorer unavailable. No duplicate, no posting on a guess	invariant	integration / staging	Ledger Posting Design 19b	M7
D-CONS1 … D-CONS28	Each rule of ERD Appendix C: a write that breaks it is refused by the mechanism named (key, CHECK, trigger or validation)	exact (refused)	database	ERD Appendix C	M2
D-POLY1 … D-POLY7	The generic trigger refuses a type-plus-id reference whose id does not exist, for each of the seven references	exact (refused)	database	ERD Appendix C, option b	M2
D-MIG	Every migration runs against last week's restored snapshot with the ledger checks green; ledger tables additive only	invariant	release	Architecture §6 schema change policy	every release
R-VIS1 … R-VISn	Visibility by field group, per table and per role, on every read, list, export, document, notification, error message and the client journal: Operations never receives REF, ECON or LEDGER fields	invariant	integration	Roles §2, §5 rule 2, 3; Architecture §5	M6
R-ACCESS	The quarterly access review report lists every account with person_ref, last sign-in and open approvals; a dormant account can be disabled and nothing is deleted	exact	integration	Operating Procedures A.10; Roles 5a	M6
O-EVID1 … O-EVID3	Evidence checklist: a collection cannot be verified without the required evidence or an authorised exception; hash missing; cash short; partner says paid while the receiver disputes	exact	UAT	Operating Procedures A.11, D.2, D.3	M10
O-C6A	Rejected payout: each of the four outcomes follows its steps and approvers and posts as Pattern F	exact	UAT	Operating Procedures C6a	M10
O-SLA	A pending item past its approval_time_limit appears on the escalation role's list and raises an alert; cutoff warnings at the configured minutes	exact	integration	Lifecycle 8a; THRESHOLD kinds	M8
G-SEC	Go-live criterion 9: an Operations user tries to see a margin or approve a settlement through screens, API, export, notification and journal: denied, nothing changed	invariant	acceptance	Cutover Approach exit criteria 9 to 13	M10
G-CONC	Criterion 10: two users edit the same deal; two Finance users approve the same settlement; a job started twice	invariant	acceptance	Cutover Approach	M10
G-OUT	Criterion 11: the application is stopped mid-collection and mid-conversion; on restart nothing lost, nothing half-posted	invariant	acceptance	Cutover Approach	M10
G-DUP	Criterion 12: the same settlement instruction submitted twice; the same collection recorded twice	invariant	acceptance	Cutover Approach	M10
G-REST	Criterion 13: restore to a point in time and recover under the runbook 8a; two sign-offs; no missing or duplicate movement	invariant	acceptance	Cutover Approach 8a	M10, rehearsed yearly
G-OPEN	Opening entries: every Section 5 check agrees exactly with the sheets as of the opening date; in-flight deals continue in the app	exact	acceptance	Cutover Approach §4; Pattern R	M11

G-FILTER	The pre-copy filter removes every flagged test project with its rows; zero test rows in production; counts and balances exclude flagged projects	exact	acceptance	Cutover Approach §3, §7	M11
G-RECON	Ten consecutive business days of the Section 5 checks inside tolerance; a reconciliation record per partner × currency × day	range (tolerance per check)	parallel run	Cutover Approach §5, §6; Pattern Q	M10
G-VULN	Basic vulnerability assessment of staging; critical and high findings fixed before production	range (severity)	security	Architecture 6a	before go-live

6 · Coverage sheet: the Workflow Tree against the tests

One row per sub-class of the Workflow Tree (27 sub-classes, 92 procedures). For each: where the procedures are written up, which client scenarios of the Scenario Catalogue use them, which ledger pattern books them, which tests exercise them and how many. A count of zero would be an open area; today none is zero, but the two lowest rows (6.2 payout methods, 7.5 ledger corrections) are where Phase 3 adds UAT cases first. The sheet is kept as a file next to the test suite (coverage.csv), with one row per procedure (the 92 leaves) filled in as the build phases land, and is shown at every phase review; the sub-class view here is its summary at the start of the build. Note: coverage.csv is a build-phase deliverable (Phase 1 onwards, maintained through Phase 3), not part of the P0 review cycle; its test column can only be filled once the test suite exists.

Workflow Tree sub-class	Written up	Client scenarios (Catalogue)	Ledger pattern	Tests	Count
1.1 Start a new project	OP Part C; CS 1a	all	—	V-ED, D-CONS (project state)	2
1.2 Change configuration	CS §1 change control, 1a	all	—	V-ED-A…D, D-CONS (write-once)	5
1.3 Parties, endpoints and destinations	CS §4, §5; OP C2, C3, C7	1.1–1.6, 2.1–2.5, 4.1–4.4	—	D-CONS (accounts, endpoints, single-use), R-VIS	4
1.4 Users, roles and access	Roles §5, 5a; OP A.10	all	—	C-APPR, R-VIS, R-ACCESS	3
2.1 Partner rate versions	Calc §2, 2a; LC Figure 3	all converting scenarios	—	S-RATE, V-RB1…3	4
2.2 Market reference	Arch §4 rate capture; CS RATE_SOURCE	all	—	C-JOBS, V-RB	2
2.3 Bank cutoff day (C9)	OP C9; CS PARTNER; LC §2	2.6, 4.1, 4.3, 6.2	K (T37)	S-DEAL (cutoff), F-T37	2
3.1 Price and send a quote	LC §2; Calc §3–§7; X1	2.1–2.5, 4.x	A	V-L1…L9, V-B1…B23, S-DEAL	28
3.2 Validity, grace and late decisions	LC §2, OP A.2, D.1	2.4	K (T38)	S-DEAL, F-T38	2
3.3 Entry modes without a quote	LC §2; SC W2	2.4, 6.1	A	S-DEAL (collect-first), V-ED-D	2
3.4 Fee mechanics decided at quote	Calc §4, 4a, §5, 5a, §6, §11	1.4, 1.5, 4.3, 5.1	J	V-TIER1…2, V-L6, V-L12, V-L13, F-T30…T35	10
4.1 Collection methods	OP C1, C2, C4, C7; X2	1.x, 2.x, 3.2, 4.2, 5.x	A, B, D	O-EVID1…3, F-T1, T4, T5, T15…T17	9
4.2 Verify the collection	OP A.3, A.11; LC §5	all	A, I	O-EVID, F-T28, T29, D-CONS (endpoint)	5
5.1 Conversion on a deal leg	LC §2, §5; Calc §8, §12; X3	all converting	A, K	F-T2, T3, T6, T11, T14, T36, S-DEAL	7
5.2 Conversions not on a deal	LC §4; Calc §16; X6	6.2	H	F-T26, T27, S-DEAL (balance conversion)	3

6.1 Prepare, approve and release	LC §3, Figure 2; OP A.5; X4	all paying out	C	F-T7…T10, S-SETT, O-SLA, G-DUP	7
6.2 Payout methods	OP C1, C3, C5, C8, C10	1.3, 1.6, 3.3, 4.4, 5.x, 6.3	C	O-EVID, F-T23	2
6.3 Bank fees and returns	OP C6, C6a, D.2, D.3; LC §3	2.7, 4.1	C, F	F-T9, T10, T22…T25, O-C6A	7
7.1 Reroute and custody	LC Figure 4, 2b; OP A.8; X5	1.x (Ali unavailable), 2.5	D, O	F-T12…T17, T56…T58, S-RR, C-INT1	11
7.2 Voids and unwinds	LC §5; OP D.4; Ledger E	—	E	F-T18…T21, S-DEAL (voids)	5
7.3 Loss events and recoveries	LC Figure 6; OP C7, D.5	4.2	L	F-T39…T43	7
7.4 Earnings resting at partners	LC Figure 7; Ledger N; X6	1.x, 2.x	N	F-T52…T55, S-RR (dues)	5
7.5 Ledger corrections	Ledger E addition, rule 11a	—	E	F-REV, F-T19	2
8.1 Daily	OP A.6, A.12; Cutover §5	all	P, Q	G-RECON, C-INT1…3, C-JOBS	6
8.2 Month end	OP A.7; Ledger §20, Pattern Q; X7	1.x, 3.3	J, M, Q	F-T31, T34, T35, T44…T51, T59, F-ID1…12	25
8.3 Reports the ledger answers	Ledger §21; Roles	all	—	R-VIS, F-ID	2
8.4 Cutover: parallel run, opening entries and switch	Cutover §3, §4, §6, §7, 8a; Pattern R	all	R	G-OPEN, G-FILTER, G-REST, G-RECON, F-T60, T61	6

7 · The two files that travel with this document

File	Content	Owner
vectors-v5.9.csv	One row per acceptance vector L1 to L13 and B1 to B23: every input (amount, rates, fee parts, mode, rounding) and every expected figure, as printed in the Calculation Specification; the automated tests read this file, so a change to a vector is a change to this file with a recorded reason	New XP; NPL Finance approves changes to B rows
coverage.csv (build phase)	One row per procedure of the Workflow Tree (92 leaves) with the columns of Section 6; counts are computed from the test suite at each phase review. Created in Phase 1 (M0) from the Workflow Tree, filled as the milestones land; not a P0 deliverable	New XP, from Phase 1

Identifiers in this document are frozen from this edition. New tests take the next free number in their family; a withdrawn test keeps its number with the note "withdrawn".